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Nitrate-to-ammonium ratio shapes vegetative growth, nutrient status, and physiology of *Vanilla planifolia* in soilless culture

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Abstract

Vanilla is a high-value crop whose demand exceeds supply, yet evidence-based nutrient management remains unexplored. This study investigated the effect of nitrogen form on vegetative growth, biomass allocation, and physiology of *Vanilla planifolia* in soilless culture. Ninety-five plants were assigned to five NO₃⁻-N: NH₄⁺-N ratios (0:100 to 100:0; $n = 17$ each) at 47 mg nitrogen L⁻¹ and a nitrogen-free solution ($n = 10$), over 140 days under controlled conditions. Total dry mass declined linearly with increasing ammonium proportion ($p < 0.001$), with relative growth rate the most responsive parameter: ammonium nutrition reduced total dry mass by 28% and relative growth rate by 36% relative to purely nitrate-fed plants. Increasing ammonium supply acidified the rhizosphere (substrate pH declined by 1.2 units, from 5.93 to 4.72) and tended to reduce leaf biomass fraction and total root length; tissue nitrogen concentration increased linearly with increasing ammonium proportion across all organs. A treatment $\times$ shoot-position interaction revealed temporally compounding growth advantages under nitrate nutrition. These results provide first evidence that nitrogen form, together with the accompanying shift in rhizosphere pH, shapes early vegetative growth in vanilla, with nitrate-dominated supply producing the highest vegetative growth under the controlled soilless conditions tested.

Keywords Soilless cultivation, Fertilisation, Hydroponic, Nitrogen, Vanilla

1 Introduction

Vanilla planifolia (*V. planifolia*) is the primary commercial source of natural vanilla, one of the most valued spices worldwide, with applications spanning the food, beverage, cosmetics, and pharmaceutical industries [1]. The global vanilla bean market is projected to grow from approximately USD 1.1 billion in 2022 to USD 1.79 billion by 2029 [2], yet global production of raw vanilla, estimated at approximately 7,114 tonnes in 2024 [3], remains insufficient to meet demand [4]. This persistent supply deficit has driven reliance on synthetic vanillin substitutes and has underscored the need for improved and diversified cultivation practices to stabilise natural vanilla production [1, 4]. As a



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semi-epiphytic CAM orchid native to Mexico, *V. planifolia* typically requires three years of vegetative growth before entering the generative phase [1, 5], making an extended, nutrient-supported vegetative period essential for establishing productive plants. Understanding the nutritional requirements during this prolonged phase therefore has direct relevance for improving vanilla cultivation.

Nitrogen (N) is the mineral nutrient required in the greatest quantity by higher plants and is a fundamental constituent of amino acids, proteins, nucleic acids, and chlorophyll [6]. Accordingly, N supply directly governs photosynthetic capacity and biomass accumulation across crop species [6, 7]. Beyond total supply, the form in which N is provided also influences root system architecture; nitrate regulates lateral root development via auxin signalling [8], whereas ammonium enhances lateral root branching [9]. The form in which N is supplied, whether as ammonium (NH_4^+), nitrate (NO_3^-), or a combination of both, is an important factor influencing plant growth and rhizosphere chemistry [6]. In soilless systems with inert substrates such as perlite, where microbial nitrification is reduced and organic nitrogen mineralisation is negligible, the applied NO_3^- -N: NH_4^+ -N ratio is more closely preserved in the root zone than in soil-based systems [10, 11], making nitrogen-form effects both more direct and more consequential. This precise control over individual nutrient factors makes soilless systems particularly suited for investigating nitrogen-form effects.

Plants differ considerably in their preference for and tolerance of the two inorganic N forms. A recent meta-analysis encompassing over 2,400 observations from 151 studies confirmed that the optimal NO_3^- -N: NH_4^+ -N ratio is species-specific and depends on both the crop and the growing conditions [12]. While many horticultural crops perform best under nitrate-dominated nutrition, the inclusion of a moderate ammonium fraction can enhance N uptake efficiency, promote root development, and support rhizosphere pH regulation [12, 13]. Conversely, when NH_4^+ constitutes the sole or dominant N source, a syndrome of ammonium toxicity may develop, characterised by rhizosphere acidification, competitive inhibition of Ca^{2+} and Mg^{2+} uptake, and impaired shoot growth [13–15].

The influence of nitrogen form on crop performance has been documented across a wide range of horticultural species grown in soilless systems. In strawberry, a 75:25 NO_3^- -N: NH_4^+ -N ratio maximised yield and photosynthetic rate, whereas ratios with 75% ammonium reduced leaf area and fruit production [16]. Similar nitrate preferences have been reported for kale [17]. Importantly, nitrogen-form effects are not static, and temporal dynamics have been documented in several species. In kale, 75% ammonium caused no adverse effects after four weeks but significant growth reductions after eight weeks [17], and in blueberry, the optimal ratio for photosynthetic pigment accumulation shifted from 25:75 (NO_3^- -N: NH_4^+ -N) to 50:50 over a 90-day period [18]. Such findings indicate that nitrogen-form effects may intensify, attenuate, or shift qualitatively with prolonged exposure.

Within the Orchidaceae, the available evidence suggests a general sensitivity to high ammonium levels. Epiphytic orchids such as *Phalaenopsis* and *Dendrobium* grow and flower best when ammonium constitutes no more than 40–50% of total N, whereas ammonium proportions exceeding 75% cause growth inhibition and toxicity symptoms after twelve months of treatment [19]. A nitrate-to-ammonium ratio of at least 3:1 has been recommended for *Phalaenopsis* to promote vigorous vegetative growth and

flowering [20]. However, these studies were conducted on potted ornamental orchids grown in bark or moss substrates, and their applicability to climbing, semi-epiphytic species such as *V. planifolia* cultivated in an inert substrate remains untested.

For *V. planifolia* specifically, knowledge of optimal nutrient management remains limited. Supplemental fertilisation has been shown to enhance vegetative growth in soil-grown vanilla [21], and to increase fruit yield from 0.55 to 1.5 kg per plant [22]. More recently, young *V. planifolia* plants have been shown to sustain biomass production in a substrate-free hydroponic system at a comparatively low electrical conductivity (EC) level of 0.9 mS cm⁻¹ using nitrate-dominated N nutrition [23]. For the closely related nothospecies (hybrid) *Vanilla × tahitensis*, six N application rates (0–32 g N plant⁻¹ yr⁻¹) supplied as urea ammonium nitrate solution (UAN, 28% N) were evaluated under controlled container conditions, yielding curvilinear growth responses with an optimum intermediate range of 13–21 g N plant⁻¹ yr⁻¹ ($\approx$ 16 g on average) and corresponding leaf N concentrations of 1.5–2.2% [24].

While the studies reviewed above indicate that vanilla benefits from nitrogen fertilisation, none have isolated the effect of nitrogen form under controlled conditions. It is therefore unknown which NO₃⁻-N: NH₄⁺-N ratio best supports vegetative growth, whether physiological vigour is compromised under ammonium-dominated nutrition, and whether such responses remain stable or shift with prolonged exposure. Resolving these questions is particularly relevant given the extended vegetative phase required before *V. planifolia* enters the generative stage, during which nitrogen management must sustain prolonged growth and plant vigour [1, 5]. To address these knowledge gaps, the objectives of this study were (i) to determine the effect of the NO₃⁻-N: NH₄⁺-N ratio in the nutrient solution on vegetative growth (biomass accumulation and morphological development) of *V. planifolia*, (ii) to assess the physiological vigour of the plants in response to nitrogen form (based on chlorophyll fluorescence, stomatal conductance, SPAD, and tissue nitrogen content), and (iii) to evaluate whether the growth and physiological response to nitrogen form differs with prolonged exposure. To our knowledge, this is the first systematic comparison of nitrogen forms in *V. planifolia* under controlled soilless conditions.

2 Materials and methods

2.1 Plant material and pre-cultivation

Thirty-centimetre shoot cuttings with three nodes were obtained from healthy *V. planifolia* specimens aged two to three years. Prior to the experiment, cuttings were grown for four months in PE pots (11 cm diameter) filled with 80 g of perlite substrate. Cultivation took place under controlled environmental conditions, maintaining a temperature of 27 ± 0.3 °C, relative humidity (RH) of 83 ± 2.5%, and a photosynthetic photon flux density (PPFD) of 63 µmol m⁻² s⁻¹ provided by Valoya LED modules (380–750 nm emission range, Finland) set to a 12-hour photoperiod. Plants received 100 mL tap water weekly without additional fertilisation (Supplementary Table S1). After pre-cultivation, healthy and morphologically uniform plants were selected for the experiment.

2.2 Experimental design and growth conditions

Ninety-five plants were assigned to six treatment groups in a randomised design comparing five NO₃⁻-N: NH₄⁺-N ratios (100:0, 75:25, 50:50, 25:75, and 0:100; $n = 17$ each) at

a nominal total nitrogen concentration of 47 mg L^{-1} , plus a nitrogen-free control group ($n = 10$; Fig. 1). The five ratios spanning the full 100:0 to 0:100 range were chosen to characterise the dose–response relationship across the nitrate–ammonium continuum rather than to match a specific commercial formulation, as vanilla-specific ratios are not established. Nitrate was supplied as calcium nitrate and potassium nitrate; ammonium was supplied as ammonium sulphate and monoammonium phosphate. All macronutrients other than nitrogen were balanced across treatments, apart from sulphate, which co-varied with the $\text{NO}_3^- \text{-N} : \text{NH}_4^+ \text{-N}$ ratio as an obligate counterion of ammonium supply. EC of inflow nutrient solution across treatments was $0.67 \pm 0.16 \text{ mS cm}^{-1}$. Nutrient solutions were formulated using established hydroponic concentration ranges adapted for vanilla [6, 10, 25, 26], with target EC and nitrogen concentration based on functional ranges established for *V. planifolia* in soilless culture [23]. This total-nitrogen level derives from these previously established functional ranges rather than a commercial standard, as evidence-based nitrogen recommendations specific to commercial vanilla are not yet available. All solutions were adjusted to pH 7 using potassium hydroxide before application. Solution pH was set before each application; leachate pH was monitored at every application and substrate pH determined at harvest, but no corrective in-culture pH adjustment was applied, to avoid imposing treatment-specific ionic loads. Detailed nutrient compositions are provided in Supplementary Table S1.

The experiment was conducted in an indoor facility using a rack system (13.4 m^2) equipped with APOLLO1 TUNEABLE multi-wavelength LED modules (FUTURELED, Germany), providing $77 \mu\text{mol m}^{-2} \text{ s}^{-1}$ total PPFD under a 12-hour photoperiod. Recorded climate conditions averaged $26.5 \pm 0.63 \text{ }^\circ\text{C}$, $73.6\% \pm 15.3 \text{ RH}$, and 505 ± 79

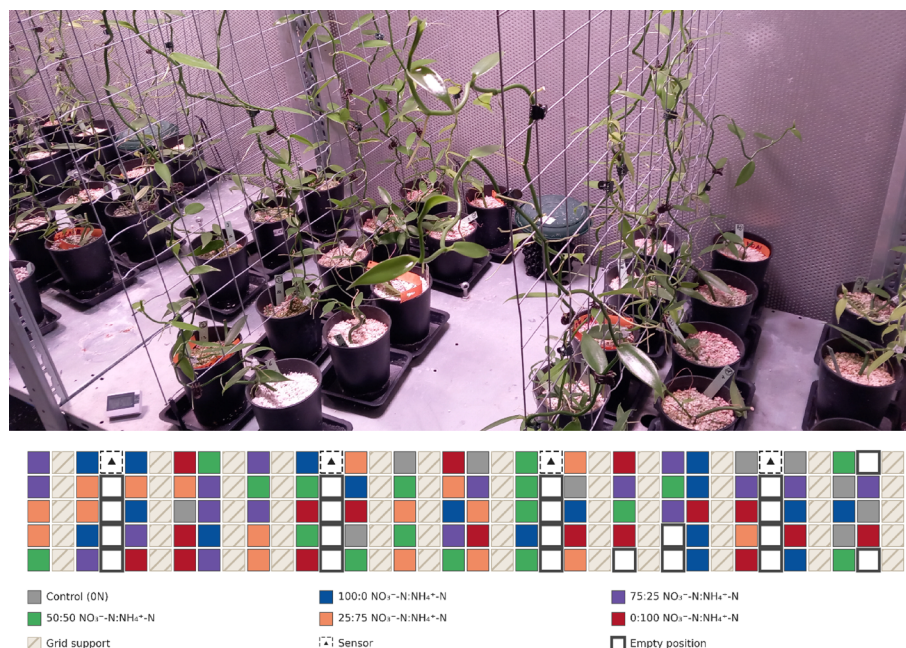


Fig. 1 Spatial arrangement of *Vanilla planifolia* plants in the randomised experimental design on the indoor rack system (5 rows with up to 36 positions per row; 13.4 m^2). Coloured cells represent individual plants assigned to one of six nitrogen-form treatments: Control (0 N; $n = 10$), 100:0 $\text{NO}_3^- \text{-N} : \text{NH}_4^+ \text{-N}$ ($n = 17$), 75:25 $\text{NO}_3^- \text{-N} : \text{NH}_4^+ \text{-N}$ ($n = 17$), 50:50 $\text{NO}_3^- \text{-N} : \text{NH}_4^+ \text{-N}$ ($n = 17$), 25:75 $\text{NO}_3^- \text{-N} : \text{NH}_4^+ \text{-N}$ ($n = 17$), and 0:100 $\text{NO}_3^- \text{-N} : \text{NH}_4^+ \text{-N}$ ($n = 17$). Total nitrogen supply was constant at 47 mg L^{-1} across all nitrogen fertilised treatments. Hatched cells represent grid support structures; dashed cells with filled triangles ($\blacktriangle$) indicate environmental sensor positions; cells with bold borders denote empty positions

ppm CO₂ recorded using LOG32TH (temperature, RH) and AirCO2ntrol 5000 (CO₂) data loggers (Dostmann Electronic, Germany). Humidity was maintained using an automated fine-mist system with demineralised water. Detailed light spectra data are provided in Supplementary Figure S1. Consistent with the shade-adapted habit of *V. planifolia*, which is sensitive to high irradiance, and to maintain continuity with the conditions under which functional EC and nutrient ranges for this species were established [23], a relatively low PPFD was applied.

2.3 Treatment application

Plants received 200 mL of their respective nutrient solution per pot fortnightly (10 applications). The experimental period comprised 154 days: a 14-day pre-treatment phase (demineralised water), followed by 10 biweekly nutrient applications, and a 14-day post-application growth period before harvest (day 154). The extended interval between applications prevented waterlogging. Leachate was collected in trays below pots and remained available for plant uptake. Pooled leachate samples (5 mL per plant per treatment) were collected 15 min after each application for pH, EC, ammonium, and nitrate measurement. Two leaching events were conducted at 70 and 112 days after treatment onset, when leachate pH had declined substantially and/or EC values had spiked, to mitigate high nutrient accumulation and pH drift. Each pot received 200 mL of demineralised water (EC < 0.02 mS cm⁻¹), the leachate was discarded, and treatment solutions were reapplied 30 min later. Trays were scrubbed with demineralised water two days before each subsequent fertigation.

2.4 Leachate and substrate analysis

Pooled leachate samples within each treatment group were assessed for EC and pH using handheld devices (PH603 + K and COND 6+; Eutech Instruments, Singapore), while ammonium and nitrate concentrations were determined using a Reflectometer RQflex® 20 (Supelco, USA). At the end of the experiment, the substrate from each pot was collected and pH and EC was determined according to VDLUFA guidelines [27] using a moist perlite analysis with 0.01 mol L⁻¹ CaCl₂ solution (EC 2.2 mS cm⁻¹).

2.5 Vegetative growth analysis

Vegetative growth was assessed at the end of the experiment by collecting mass and morphometric data from each individual plant. Plant organs were measured separately for roots, shoots, and leaves. Shoot and leaf measurements were further divided into apical (developed during the second half of the treatment period) and basal (developed during the first half) positions to assess positional growth responses along the shoot axis. Dry mass (DM) was determined by oven-drying tissue samples to constant weight at 60 °C following [28]. Morphometric parameters were quantified using digital imagery processed with ImageJ/Fiji software (version 1.54p; [29]). From primary measurements, total dry mass (total DM), root mass fraction (RMF), leaf mass fraction (LMF), root-to-shoot ratio (R: S), specific leaf area (SLA), and relative growth rate (RGR) were calculated. For RMF, LMF and R: S, total DM included the plant (initial cutting) base, which was partitioned into shoot and leaf fractions based on the mean organ distribution of the three intermediate NO₃⁻-N: NH₄⁺-N treatments. RGR was calculated from aboveground

dry mass only, because initial root dry mass could not be determined non-destructively (formulas in Supplementary Table S2).

2.6 Physiological measurements

Relative chlorophyll content (SPAD) was measured at apical and basal leaf positions (leaf tip and leaf centre) and averaged per position. Stomatal conductance and maximum quantum efficiency of photosystem II (Fv/Fm) were recorded as whole-plant measurements. Detailed measurement protocols are provided in Supplementary Table S2.

2.7 Elemental analysis of plant tissues

Total nitrogen content was determined on a DM basis using the Dumas combustion method (DIN EN ISO 16634-2:2016-11). All analyses were performed by SGS Analytics Germany GmbH (Jena). For nitrogen, organ-separated pooled samples (roots, shoots, leaves) were prepared by combining five to six plants per replicate, yielding three biological replicates per organ and treatment for the nitrogen-fertilised groups ($n=3$) and two for the unfertilised control ($n=2$). For one pooled leaf sample per nitrogen-fertilised treatment ($n=5-6$), concentrations of P, K, Ca, Mg, S, Fe, Cu, Zn, B, and Na were determined by ICP-MS (DIN EN ISO 17294-2:2024-12) after HNO_3 pressure digestion (DIN EN 13805:2014-12; SGS Analytics Germany GmbH, Jena). Both the nitrogen (Dumas) and multi-element (ICP-MS) determinations were carried out by a laboratory accredited to DIN EN ISO/IEC 17,025 (SGS Analytics Germany GmbH, Jena) for these methods in agricultural biological materials, ensuring validated procedures, traceable calibration, and internal quality control.

2.8 Statistical analysis

All analyses were performed in R (v 4.5.2; R Core Team, 2025) at $\alpha=0.05$. Residual normality (Shapiro-Wilk) and variance homogeneity (Levene's test) were assessed prior to each analysis. Where assumptions were met, treatment effects on whole-plant parameters were tested using one-way ANOVA with Tukey's HSD post-hoc comparisons. Right-skewed parameters (root-to-shoot ratio, specific leaf area) were log-transformed before analysis. Stomatal conductance, for which neither transformation yielded satisfactory residuals, was analysed using a Kruskal-Wallis test with Dunn's post-hoc comparisons (Benjamini-Hochberg correction). Total root length, showing normally distributed residuals but unequal group variances, was analysed with Welch's ANOVA and Games-Howell comparisons. Substrate pH and EC, both exhibiting severely non-normal residuals, were analysed using Kruskal-Wallis tests with Dunn's comparisons (Bonferroni correction).

Parameters measured at apical and basal shoot positions (shoot dry mass, leaf dry mass, shoot length, total leaf area, and relative chlorophyll content) were analysed using linear mixed-effects models with treatment, position, and their interaction as fixed effects and plant identity as a random intercept. Total leaf area was log-transformed to meet model assumptions. Type III ANOVA was applied, and estimated marginal means were compared using Tukey-adjusted contrasts.

Where treatment effects were significant, the response shape along the ammonium gradient was explored using polynomial regression with NO_3^- -N proportion (0-100%) as a continuous predictor, excluding the nitrogen-free control. For significant

treatment-by-position interactions, regressions were fitted separately by organ position. Model selection followed a stepwise approach: the quadratic term was retained if significant ($p \leq 0.05$); otherwise, a linear model was fitted. All regressions used heteroscedasticity-consistent standard errors (HC3). Tissue nitrogen concentration (% DM) was regressed against NO_3^- -N proportion separately for roots, shoots, and leaves; for leaf tissue, where residual normality was violated by a single outlier, Spearman rank correlation confirmed the monotonic relationship. Group differences are denoted by lowercase letters from post-hoc comparisons. Full details on test selection criteria, assumption diagnostics, and results are provided in Supplementary Table S3.

3 Results

3.1 Root-zone environment

Substrate pH after 154 days of treatment differed significantly among NO_3^- -N: NH_4^+ -N ratios (Kruskal-Wallis, $p < 0.001$; Fig. 2A). Treatments with a high nitrate proportion maintained the highest, mildly acidic pH values: the 100:0 nitrate treatment showed a mean substrate pH of 5.93 ± 0.34 , not significantly different from the nitrogen-free control (5.74 ± 0.23). Increasing the ammonium share progressively acidified the substrate, with the 75:25, 50:50, and 25:75 treatments yielding pH values of 5.39 ± 0.20 , 5.02 ± 0.31 , and 4.98 ± 0.31 , respectively. The pure ammonium treatment resulted in the lowest pH (4.72 ± 0.33), representing an overall decline of approximately 1.2 pH units from the most nitrate-rich to the most ammonium-rich treatment.

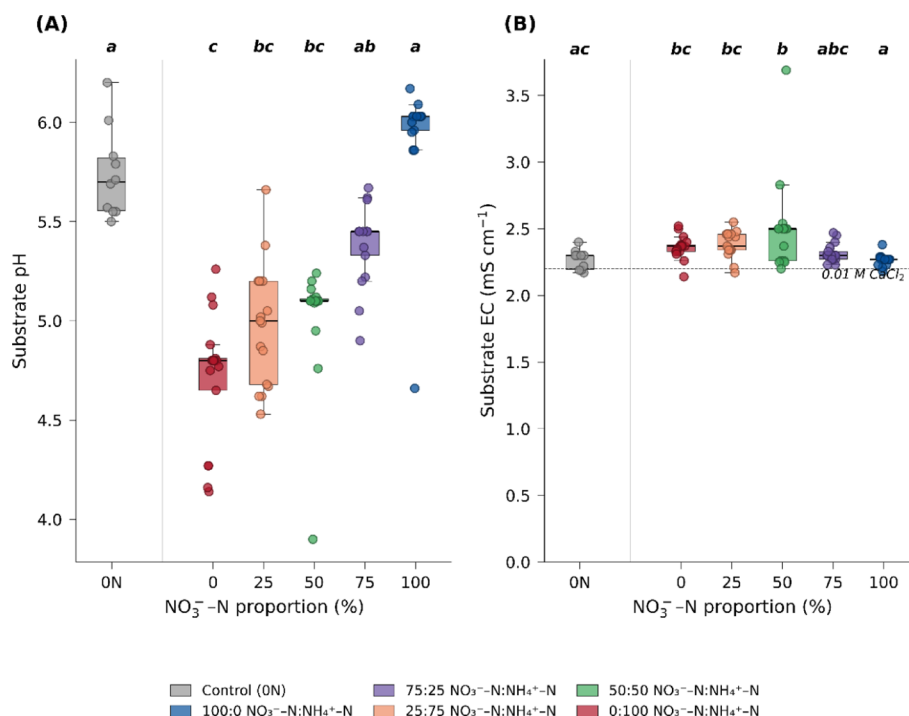


Fig. 2 Substrate pH (A) and electrical conductivity (EC) (B) of *Vanilla planifolia* grown under different NO_3^- -N: NH_4^+ -N ratios, measured at the end of the 154-day cultivation period (140 days after the onset of nitrogen treatments). Boxplots display the median (horizontal line), interquartile range (box), and 1.5 × IQR (whiskers); individual data points are shown as jittered dots. The dashed line in (B) indicates the background EC of the 0.01 M CaCl_2 extraction solution (2.2 mS cm⁻¹). The control (0 N) received a nitrogen-free solution. Treatments sharing the same lowercase letter are not significantly different (Kruskal-Wallis test followed by Dunn's post-hoc test with Bonferroni correction, $\alpha = 0.05$). $n = 17$ per nitrogen treatment, $n = 10$ for Control (0 N)

Substrate EC also differed significantly among treatments (Kruskal-Wallis, $p < 0.001$; Fig. 2B); however, all group means remained close to the background EC of the 0.01 M CaCl_2 extraction solution (2.2 mS cm^{-1}), indicating that treatment-driven variation was small.

Leachate pH tended to decline over the course of the experiment in the ammonium-dominated treatments (25:75 and 0:100), whereas the nitrate-dominated treatments (100:0, 75:25) maintained relatively stable leachate pH values throughout the monitoring period (Fig. 3A). Leachate EC showed an initial rise in the ammonium-rich treatments followed by partial recovery after the two leaching events at 70 and 112 DAT (Fig. 3B). Nitrate-nitrogen recovery in the leachate appeared highest in the 100:0 treatment and declined with decreasing nitrate proportion in the nutrient solution, while ammonium-nitrogen in the leachate showed the inverse pattern (Fig. 3C, D). These temporal patterns are consistent with the substrate pH gradient observed at harvest and suggest progressive acidification under ammonium-dominated nutrition, though the absence of replication precludes formal statistical confirmation ($n = 1$).

3.2 Growth and physiological responses to nitrogen form

Nitrogen form affected total DM and RGR of *V. planifolia* ($p < 0.001$; Table 1). All nitrogen-supplied plants accumulated more total DM than the nitrogen-free control, except

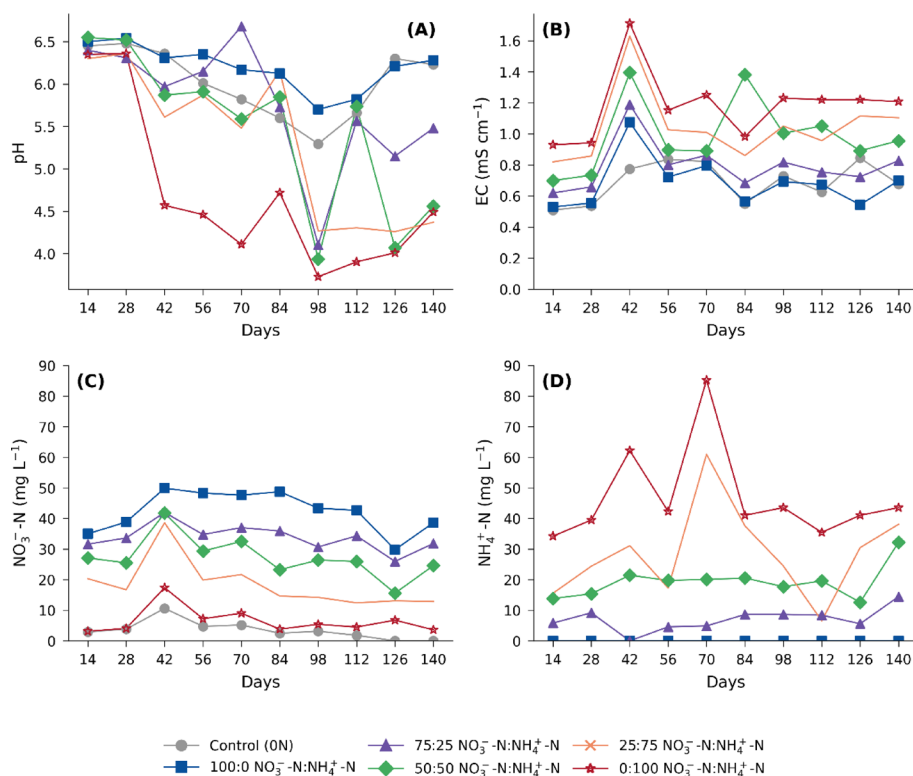


Fig. 3 Leachate pH (A), electrical conductivity (EC) (B), nitrate-nitrogen (NO_3^- -N) (C), and ammonium-nitrogen (NH_4^+ -N) (D) of *Vanilla planifolia* grown in perlite under six nitrogen-form treatments over the 140-day treatment period (leachate sampled at each of 10 biweekly applications). Treatments comprised different NO_3^- -N: NH_4^+ -N ratios at a constant total nitrogen supply of 47 mg L^{-1} : Control (0 N, no nitrogen), 100:0, 75:25, 50:50, 25:75, and 0:100 NO_3^- -N: NH_4^+ -N. Nutrient solution (200 mL) was applied biweekly (10 applications). Each data point represents a single pooled measurement per treatment per time point ($n = 1$). The transient EC peak at day 42 reflects incomplete rinsing of pot underlays prior to leachate collection at that sampling point and is not indicative of a treatment effect

Table 1 Effect of the $\text{NO}_3^-:\text{N}:\text{NH}_4^+:\text{N}$ ratio on vegetative growth, biomass allocation, and physiological parameters of *Vanilla planifolia* grown in perlite over 154 days.

Parameter	Control (0 N)	100:0 $\text{NO}_3^-:\text{N}:\text{NH}_4^+:\text{N}$	75:25 $\text{NO}_3^-:\text{N}:\text{NH}_4^+:\text{N}$	50:50 $\text{NO}_3^-:\text{N}:\text{NH}_4^+:\text{N}$	25:75 $\text{NO}_3^-:\text{N}:\text{NH}_4^+:\text{N}$	0:100 $\text{NO}_3^-:\text{N}:\text{NH}_4^+:\text{N}$	η^2	p
Total dry mass (g)	0.89 ± 0.24 ^c	1.73 ± 0.35 ^a	1.68 ± 0.47 ^{ab}	1.57 ± 0.52 ^{ab}	1.50 ± 0.44 ^{ab}	1.25 ± 0.41 ^{bc}	0.281	***
RMF (g g ⁻¹) †	0.305 ± 0.070	0.244 ± 0.048	0.248 ± 0.053	0.255 ± 0.054	0.253 ± 0.060	0.267 ± 0.044	0.093	ns
LMF (g g ⁻¹) †	0.280 ± 0.030 ^b	0.333 ± 0.032 ^a	0.332 ± 0.026 ^a	0.321 ± 0.043 ^a	0.326 ± 0.037 ^a	0.304 ± 0.025 ^{ab}	0.226	***
R: S ratio (g g ⁻¹) †	0.45 ± 0.15	0.33 ± 0.09	0.34 ± 0.10	0.35 ± 0.11	0.35 ± 0.11	0.37 ± 0.09	0.096	ns
SLA (cm ² g ⁻¹) †	132.63 ± 42.19 ^{ab}	94.41 ± 24.84 ^b	92.62 ± 23.46 ^b	101.72 ± 23.21 ^{ab}	109.77 ± 50.86 ^{ab}	138.06 ± 61.81 ^a	0.160	**
Fv/Fm	0.675 ± 0.067 ^b	0.739 ± 0.040 ^a	0.727 ± 0.051 ^{ab}	0.732 ± 0.039 ^a	0.763 ± 0.058 ^a	0.734 ± 0.032 ^a	0.201	**
gs (mmol m ⁻² s ⁻¹)	21.53 ± 4.80 ^a	23.51 ± 5.30 ^a	25.73 ± 9.15 ^a	33.34 ± 44.61 ^a	24.06 ± 8.36 ^a	23.83 ± 8.52 ^a	0.036	ns
Total root length (cm)	159.05 ± 36.64 ^{ab}	222.93 ± 82.74 ^a	205.85 ± 84.76 ^{ab}	209.86 ± 71.08 ^{ab}	201.82 ± 67.08 ^{ab}	143.53 ± 41.37 ^b	0.153	**
Total Leaf area (cm ²)	22.31 ± 11.00 ^b	49.21 ± 12.93 ^a	45.33 ± 13.51 ^a	44.61 ± 19.45 ^a	46.90 ± 23.08 ^a	38.85 ± 11.75 ^{ab}	0.193	**
RGR aboveground (g g ⁻¹ d ⁻¹)	0.0041 ± 0.0014 ^d	0.0095 ± 0.0016 ^a	0.0084 ± 0.0013 ^{ab}	0.0078 ± 0.0018 ^{bc}	0.0077 ± 0.0017 ^{bc}	0.0061 ± 0.0016 ^c	0.508	***

Whole-plant parameters were analysed by one-way ANOVA with Tukey HSD post hoc test, except stomatal conductance (Kruskal–Wallis with Dunn's test, Benjamini–Hochberg correction) and total root length (Welch's ANOVA with Games–Howell test). Values are means ± SD; different superscript letters denote significant differences among treatments ($\alpha=0.05$). Effect size is η^2 throughout; for stomatal conductance the rank-based η^2 ($\eta^2_{\text{r-H}}$) is given. $n=17$ per N treatment and 10 for Control 0 N. †Log-transformed for analysis; untransformed values shown. *** $p<0.001$, ** $p<0.01$, * $p<0.05$, ns=not significant. For detailed sample sizes, exclusions, and full test statistics, see Supplementary Table S3.

those receiving exclusively ammonium. Among the nitrogen treatments, purely nitrate-fed plants (100:0) produced the highest total DM (1.73 g), surpassing the ammonium-only treatment, while intermediate ratios did not differ from either extreme. RGR showed the strongest treatment effect of all parameters measured ($\eta^2 = 0.508$) and revealed finer resolution among the nitrogen treatments. Plants supplied with nitrate only ($0.0095 \text{ g g}^{-1} \text{ d}^{-1}$) grew faster than all treatments containing $\geq 50\%$ ammonium, while the 75:25 (NO_3^- -N: NH_4^+ -N) ratio sustained growth rates statistically similar to pure nitrate yet significantly exceeded ammonium-only nutrition. Notably, plants grown under exclusive ammonium supply ($0.0061 \text{ g g}^{-1} \text{ d}^{-1}$) achieved the lowest RGR numerically of any nitrogen treatment, significantly below both the pure nitrate and 75:25 treatments.

Nitrogen supply promoted leaf investment relative to the nitrogen-free control. LMF was significantly higher in all nitrogen treatments except the ammonium-only treatment, which remained intermediate (Table 1). Neither RMF nor R: S ratio differed significantly among treatments. Ammonium-dominated nutrition produced leaves with greater specific area (Table 1), with SLA highest in the 0:100 treatment ($138.06 \text{ cm}^2 \text{ g}^{-1}$), exceeding the 100:0 and 75:25 treatments, while the remaining treatments did not differ from either group. Total leaf area was higher in all nitrogen treatments than in the nitrogen-free control except the 0:100 (NO_3^- -N: NH_4^+ -N) treatment, which overlapped with the control, though no differences emerged among nitrogen treatments themselves. Total root length followed a similar pattern to total DM, peaking under pure nitrate (222.93 cm) and falling to its lowest under pure ammonium (143.53 cm), with intermediate ratios overlapping both. Fv/Fm was reduced in the nitrogen-free control (0.675) relative to all nitrogen treatments except the 75:25 treatment but did not differ among nitrogen forms (range 0.727–0.763). Stomatal conductance was unaffected by the nitrogen solution applied.

For the five positional parameters analysed by linear mixed models, treatment and the treatment $\times$ position interaction reached significance for all five parameters; the position main effect was significant for all except total leaf area (Table 2). Across nitrogen-supplied treatments, shoot and leaf DM were higher in apical than in basal organs, reflecting the acropetal growth habit of the vine, but the magnitude of this divergence varied with nitrogen form ($p < 0.001$ for both Treatment $\times$ Position interactions). In the 100:0 (NO_3^- -N: NH_4^+ -N) treatment, apical shoot DM exceeded basal by 48%, whereas in the 0:100 treatment apical and basal values converged. A similar pattern of diminishing apical-basal divergence with increasing ammonium proportion was observed for shoot length and total leaf area ($p < 0.01$ for both interactions). SPAD values were consistently higher in basal than in apical leaves, and this gradient interacted with nitrogen form ($p < 0.05$). The gradient was widest at both ends of the nitrogen-form range, with basal SPAD exceeding apical SPAD by 28% under pure nitrate and by 32% under pure ammonium, and narrowest at intermediate ratios (11–16%). Averaged across positions, SPAD was lower in the 100:0 treatment than in the 75:25, 25:75 and 0:100 (NO_3^- -N: NH_4^+ -N) treatments.

3.3 Dose-response relationships along the nitrogen-form gradient

Total DM increased linearly with increasing NO_3^- -N proportion ($R^2_{\text{adj}} = 0.108$, $p < 0.001$). This relationship was more pronounced for RGR ($R^2_{\text{adj}} = 0.290$, $p < 0.001$),

Table 2 Effect of the NO₃⁻:N:NH₄⁺-N ratio on shoot-positional (apical and basal) parameters of *Vanilla planifolia* grown in perlite over 154 days, analysed by linear mixed models.

Parameter	Control (0 N)	100:0 NO ₃ ⁻ :N:NH ₄ ⁺ -N	75:25 NO ₃ ⁻ :N:NH ₄ ⁺ -N	50:50 NO ₃ ⁻ :N:NH ₄ ⁺ -N	25:75 NO ₃ ⁻ :N:NH ₄ ⁺ -N	0:100 NO ₃ ⁻ :N:NH ₄ ⁺ -N	η ²	p	p(TxP)
Shoot DM, apical (g)	0.12 ± 0.05 ^e	0.40 ± 0.10 ^a	0.38 ± 0.14 ^{ab}	0.35 ± 0.14 ^{bc}	0.31 ± 0.10 ^c	0.24 ± 0.10 ^d			
Shoot DM, basal (g)	0.17 ± 0.05 ^e	0.27 ± 0.09 ^a	0.26 ± 0.12 ^{ab}	0.25 ± 0.11 ^{bc}	0.25 ± 0.11 ^c	0.24 ± 0.10 ^d	0.139 ±	***	***
Leaf DM, apical (g)	0.10 ± 0.03 ^d	0.33 ± 0.07 ^a	0.33 ± 0.13 ^a	0.28 ± 0.12 ^b	0.27 ± 0.11 ^b	0.18 ± 0.08 ^c			
Leaf DM, basal (g)	0.06 ± 0.04 ^d	0.20 ± 0.05 ^a	0.19 ± 0.06 ^a	0.17 ± 0.07 ^b	0.18 ± 0.08 ^b	0.15 ± 0.09 ^c	0.182 ±	***	***
Shoot length, apical (cm)	16.37 ± 7.25 ^d	41.83 ± 8.52 ^{ab}	43.38 ± 9.46 ^a	40.38 ± 8.38 ^{bc}	37.00 ± 5.37 ^c	33.87 ± 13.46 ^e			
Shoot length, basal (cm)	23.56 ± 8.00 ^d	37.35 ± 6.89 ^{ab}	36.99 ± 9.71 ^a	32.23 ± 9.89 ^{bc}	31.48 ± 7.57 ^c	32.51 ± 8.25 ^c	0.203 ±	***	**
Total Leaf area, apical (cm ²) †	9.09 ± 3.72 ^d	26.36 ± 7.68 ^a	22.21 ± 6.99 ^{ab}	24.24 ± 9.86 ^{bc}	23.15 ± 10.44 ^{ab}	17.54 ± 7.48 ^c			
Total Leaf area, basal (cm ²) †	13.22 ± 7.92 ^d	22.85 ± 7.09 ^a	23.12 ± 8.00 ^{ab}	20.37 ± 11.10 ^{bc}	23.75 ± 15.34 ^{ab}	21.31 ± 8.21 ^c	0.146 ±	***	**
SPAD, apical	9.58 ± 2.54 ^e	23.63 ± 2.76 ^b	27.16 ± 3.13 ^a	26.15 ± 4.56 ^{ab}	27.44 ± 3.84 ^a	24.77 ± 4.55 ^a			
SPAD, basal	13.62 ± 3.58 ^e	30.27 ± 3.33 ^b	30.25 ± 3.54 ^a	30.38 ± 4.57 ^{ab}	31.57 ± 4.38 ^a	32.72 ± 5.15 ^a	0.396 ±	***	*

Treatments comprised five NO₃⁻:N:NH₄⁺-N ratios (100:0, 75:25, 50:50, 25:75, and 0:100) at a constant total nitrogen supply of 47 mg L⁻¹, plus a nitrogen-free control (0 N); n = 17 per nitrogen treatment and n = 10 for the Control (0 N). Values are means ± SD; different superscript letters denote significant differences among treatment (Treatment main effect, estimated marginal means pooled across positions; α = 0.05). Effect size: partial η² for the Treatment main effect (‡), p = Treatment main effect; p(TxP) = Treatment × Position interaction; the Position main effect is reported in Supplementary Table 53. †Log-transformed for analysis; untransformed values shown. *** p < 0.001, ** p < 0.01, * p < 0.05. For detailed sample sizes, exclusions, and full test statistics, see Supplementary Table 53

the parameter most strongly influenced by nitrogen form. LMF decreased linearly with increasing ammonium proportion ($R^2_{\text{adj}} = 0.063$, $p < 0.05$), indicating reduced leaf biomass allocation under ammonium-dominated nutrition. Although the adjusted R^2 values are modest, reflecting substantial individual-level variability inherent to plant growth, the consistently significant p -values confirm that nitrogen form exerts a real, directional influence on these parameters (Fig. 4A–C).

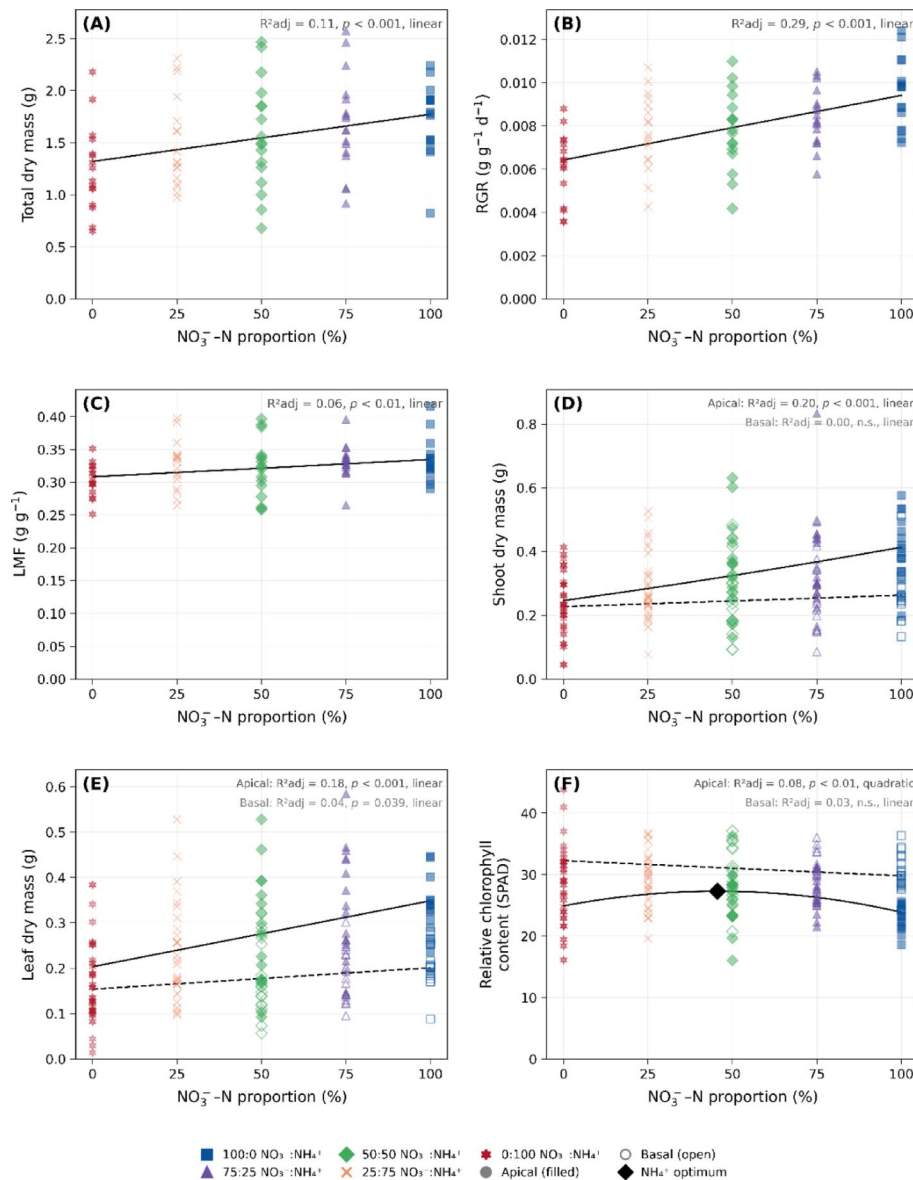


Fig. 4 Dose-response relationships between NO_3^- -N proportion (%) and growth, allocation, and physiological parameters in *Vanilla planifolia*. **(A)** Total dry mass, **(B)** relative growth rate (RGR), **(C)** leaf mass fraction (LMF), **(D)** shoot dry mass, **(E)** leaf dry mass, **(F)** relative chlorophyll content (SPAD). In panels **A–C**, a single regression was fitted across all nitrogen-supplied treatments; in panels **D–F**, separate regressions were fitted for apical (solid line, filled markers) and basal (dashed line, open markers) organs. Models were fitted as second-order polynomials with HC3-robust standard errors and simplified to linear where the quadratic term was not significant ($p > 0.05$). Shoot DM was sqrt-transformed and LMF was log-transformed prior to regression fitting; back-transformed predictions are shown. The $\blacklozenge$ symbol marks the estimated ammonium optimum where applicable. Points are coloured by treatment ($n = 83$ – 84 per panel). Adjusted R^2 and model type are annotated on each panel

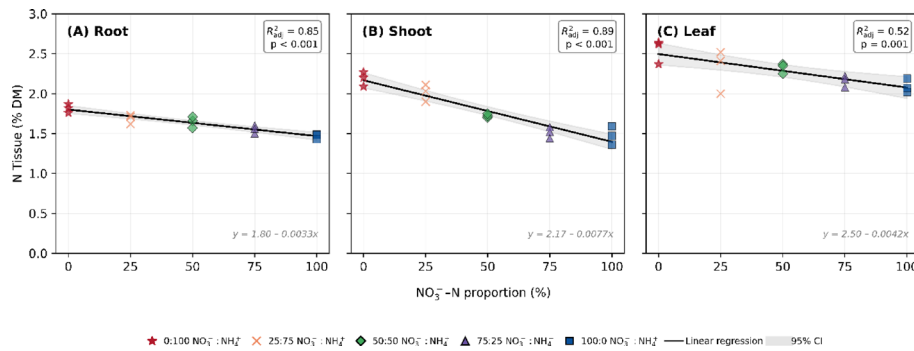


Fig. 5 Tissue nitrogen concentration (% DM) as a function of NO_3^- -N proportion (%) in (A) root, (B) shoot, and (C) leaf tissues of *Vanilla planifolia* grown under five NO_3^- -N: NH_4^+ -N ratios. The nitrogen-free control was excluded from regression. Linear models ($Y = \beta_0 + \beta_1 x$) were fitted by OLS after confirming linearity (non-significant quadratic term, $p > 0.05$). Solid lines represent fitted regressions; shaded bands indicate 95% confidence intervals. Equations, adjusted R^2 , and p -values are annotated on each panel. Data points are coloured by treatment. Total nitrogen supply was constant at 47 mg L^{-1} across all treatments. $n = 15$ per organ (3 pooled-plant replicates $\times$ 5 treatments); nutrient data are expressed on a dry-mass basis

Positional parameters revealed divergent responses between apical and basal organs. Apical shoot DM declined with increasing ammonium proportion ($R^2_{\text{adj}} = 0.197$, $p < 0.001$), whereas basal shoot DM remained unaffected. Apical leaf DM followed a similar pattern ($R^2_{\text{adj}} = 0.181$, $p < 0.001$), although basal leaf DM showed only a weak relationship ($R^2_{\text{adj}} = 0.041$, $p = 0.036$). Notably, apical SPAD values followed a quadratic rather than a linear response ($R^2_{\text{adj}} = 0.078$, $p_{\text{quadratic}} = 0.007$), with the predicted optimum at approximately 54.5% NH_4^+ -N (i.e., an approximate NO_3^- -N: NH_4^+ -N ratio of 45:55). By contrast, basal SPAD showed no such response ($p = 0.068$), indicating that the position-dependent chlorophyll optimum was confined to the actively growing apical portion of the shoot (Fig. 4D–F). As with whole-plant parameters, the low R^2 values reflect considerable plant-to-plant variation, yet the significant regression coefficients consistently support a position-dependent sensitivity to nitrogen form, particularly in the apical shoot.

3.4 Tissue nutrient composition

Root tissue nitrogen concentration was strongly predicted by the nitrate-to-ammonium balance in the nutrient solution ($R^2_{\text{adj}} = 0.848$, $p < 0.001$), with a y -intercept of 1.80% DM (corresponding to exclusively ammonium nutrition) and a decline of 0.0033% points for each additional % of NO_3^- -N supplied (Fig. 5A). Shoot tissue exhibited the strongest and most consistent response among the three organs ($R^2_{\text{adj}} = 0.889$, $p < 0.001$), with an intercept of 2.17% DM and the steepest decline of 0.0077% points per % NO_3^- -N (Fig. 5B). Leaf tissue showed a qualitatively similar but weaker response ($R^2_{\text{adj}} = 0.524$, $p = 0.001$), with the highest intercept at 2.50% DM and a decline of 0.0042% points per % NO_3^- -N (Fig. 5C). Across all organs, replacing nitrate with ammonium led to greater nitrogen accumulation per unit DM.

Unlike the tissue-nitrogen data, which rest on three pooled biological replicates per organ and treatment ($n = 3$) and were statistically evaluated, the following ICP-MS results for leaf tissue derive from a single pooled sample per treatment ($n = 1$); they are therefore presented as descriptive observations that were not statistically tested and should be interpreted cautiously until confirmed by replicated analyses. These data indicated that leaf Ca and Mg concentrations tended to decline with increasing ammonium

proportion (Ca: 2.44 to 1.42% DM; Mg: 0.53 to 0.35% DM), suggestive of ammonium-competition for uptake. Cu showed a similar declining pattern (7.87 to 6.08 mg kg⁻¹ DM). S showed the opposite trend, increasing from 0.32 to 0.52% DM, reflecting the higher sulphate supply required as counter-ion in ammonium-containing nutrient solutions. K also increased with ammonium proportion (2.65 to 3.27% DM); however, this trend may be confounded by the higher KOH volumes needed to adjust all solutions to a comparable pH before each application. The remaining macro- and micronutrients (P, Fe, Zn, B, Na) showed no clear directional response to nitrogen form (Supplementary Figure S2).

4 Discussion

4.1 Vegetative growth response to nitrogen form

The linear decline in total DM and RGR with increasing ammonium proportion places *V. planifolia* among the species that perform best under nitrate-dominated nutrition. This pattern is broadly consistent with the orchid nitrogen-form literature. In *Phalaenopsis*, nitrate proportions of at least 50% are required for satisfactory vegetative growth and flowering, and a 75:25 NO₃⁻-N: NH₄⁺-N ratio has been recommended [20]. In the present study, the 75:25 treatment sustained growth rates comparable to pure nitrate, consistent with this threshold. Ammonium proportions exceeding 75% have been reported to induce toxicity symptoms in *Phalaenopsis* and *Dendrobium*, including growth retardation, chlorotic leaves, and reduced cation accumulation [19]. Similarly, *V. planifolia* under exclusive ammonium supply showed the lowest RGR among nitrogen treatments and did not differ from nitrogen-deprived plants, suggesting growth inhibition rather than outright toxicity.

The response of *V. planifolia* stands in contrast to that of *Erythralium scandens*, a tropical woody vine that grew best under pure ammonium nutrition [30], and to centipede grass, which showed optimal biomass at a 50:50 NO₃⁻-N: NH₄⁺-N ratio [31]. Among crops grown in soilless systems, a general preference for nitrate dominance has been reported for strawberry [16, 32], kale [17], and Moldavian balm [33], all consistent with the present findings for vanilla.

An important consideration is that the observed effects cannot be attributed solely to the nitrogen form itself. Because ammonium uptake drives rhizosphere acidification [6], the growth decline under ammonium-dominated nutrition reflects the combined influence of nitrogen form, progressive substrate acidification, and their secondary consequences for nutrient availability. Because ammonium was supplied largely as ammonium sulphate, both sulphate and, correspondingly, leaf sulphur co-varied with the ammonium proportion (SO₄-S rising roughly 23-fold, from 5 to 114 mg L⁻¹). This additional load barely altered total ionic strength (inflow EC 0.67 ± 0.16 mS cm⁻¹ across treatments), however, and sulphate is a comparatively inert counter-ion for which no growth-limiting mechanism comparable to the documented pH and ammonium effects is established at these concentrations. Leaf sulphur (0.32 to 0.52% DM; single pooled samples) remained at the upper end of the range typical for crop foliage and below toxic concentrations [6]. Sulphate co-variation is therefore relevant mainly to the observed tissue-sulphur trend rather than acting as a primary driver of the growth decline, for which root-zone pH remains the dominant secondary factor. No corrective pH adjustment was applied, as any pH-correcting agent would introduce treatment-specific ion loads (e.g. Na⁺ from

NaOH, Ca^{2+} from $\text{Ca}(\text{OH})_2$, creating a secondary confound. This confounding is common in nitrogen-form studies conducted without continuous pH correction [17, 30, 33]. Nevertheless, from a horticultural standpoint, the integrated system response is the practically relevant outcome, as growers selecting nitrogen sources will encounter these coupled effects in their production systems.

4.2 Root-zone acidification and nutrient availability

The 1.2-unit decline in substrate pH from the 100% nitrate to the 100% ammonium treatment reflects proton extrusion during ammonium uptake [6], consistent with acidification reported in *Erythronium scandens* [30] and kale [17]. The minimal buffering capacity of perlite, combined with the fortnightly fertigation interval necessitated by the waterlogging sensitivity of *V. planifolia*, likely amplified this acidification compared to systems with daily nutrient delivery.

Although unreplicated ($n = 1$), the declining leaf Ca and Mg concentrations with increasing ammonium proportion are consistent with the well-documented ammonium-antagonism: NH_4^+ competes with divalent cations at root uptake sites, while proton-driven displacement reduces their availability on exchange complexes [6, 34], a pattern also reported for *Phalaenopsis* and *Dendrobium* [19]. Beyond these root-zone effects, ammonium assimilation itself carries a metabolic cost: its incorporation via the GS/GOGAT pathway draws on carbon skeletons and ATP, while the futile transmembrane ammonium cycling sustained under high external supply imposes a substantial additional energy demand [14]. Intracellular ammonium accumulation further perturbs cytosolic pH and ion homeostasis [13, 15, 35]. These cellular constraints complement the rhizosphere acidification and cation antagonism described above as contributing drivers of the growth decline under ammonium-dominated nutrition.

4.3 Biomass allocation

Nitrogen form did not alter proportional biomass allocation between roots and shoots: neither root mass fraction nor root-to-shoot ratio differed among treatments, and leaf mass fraction responded more strongly to nitrogen supply than to nitrogen form. Instead, nitrogen form affected organ morphology. Total root length declined from 222.93 cm under pure nitrate to 143.53 cm under pure ammonium, consistent with NH_4^+ -induced inhibition of root elongation [15, 36]. Conversely, specific leaf area was highest in the 0:100 treatment, indicating thinner or less dense leaves, as has been linked to altered mesophyll development under ammonium dominance in other species [37]. Together, these results suggest that ammonium stress affected organ development rather than carbon partitioning between above- and belowground compartments. The elevated SLA under exclusive ammonium may reflect reduced cell wall investment under mild ammonium stress or, alternatively, luxury nitrogen accumulation without proportional structural carbon investment. The higher tissue nitrogen concentrations in ammonium-fed plants despite lower total dry mass, a pattern consistent with luxury uptake of NH_4^+ widely documented across crop species [12, 14], supports the latter interpretation. However, ammonium toxicity itself, including direct cellular damage from excess NH_4^+ accumulation and pH-mediated nutrient imbalances [13, 35], cannot be excluded as a contributing driver. Luxury uptake and toxicity may therefore operate concurrently under high-ammonium conditions. Importantly, this elevated tissue nitrogen does not

indicate improved nitrogen nutrition: because concentrations are expressed per unit dry mass, the reduced growth under ammonium concentrates nitrogen in a smaller biomass (a reduced growth-dilution, i.e. concentration, effect), so higher tissue %N reflects lower nitrogen-use efficiency rather than superior nutrition [6, 38]. Active luxury uptake and this passive concentration effect are complementary rather than alternative and likely operate together under ammonium-dominated nutrition. The reduced total root length under ammonium also provides less absorptive surface, which may have constrained whole-plant acquisition of water and of divalent cations such as Ca and Mg [6], with potential consequences for long-term establishment. Because tissue values are expressed as concentrations rather than total contents, however, this surface-area effect acts on total uptake rather than necessarily lowering tissue concentrations and is therefore a plausible secondary contributor rather than a demonstrated cause.

4.4 Physiological vigour

Nitrogen free control plants exhibited the lowest Fv/Fm value (0.675), indicative of nitrogen deficiency impairing PSII function; similarly, nitrogen deprivation in rice has been shown to reduce Fv/Fm by increasing F_0 and decreasing Fm, reflecting damage to the entire photoelectron transport chain from the PSII donor side to the PSI acceptor side [38]. All nitrogen-fertilised treatments maintained values above 0.72, although the 75:25 (NO_3^- -N: NH_4^+ -N) treatment did not differ statistically from the control. This indicates that none of the ammonium concentrations applied here caused direct photoinhibition, and that the observed growth reductions under ammonium nutrition are more likely attributable to root-zone acidification and associated nutrient imbalances than to impairment of the photosynthetic apparatus itself. Stomatal conductance did not differ among treatments; however, as *V. planifolia* is an obligate CAM plant and measurements were taken during the day (CAM Phase III, when stomata are largely closed), this absence of differences may reflect CAM-regulated stomatal behaviour rather than a true lack of nitrogen-form effects on gas exchange. Together with the luxury uptake pattern discussed above, these findings suggest that the growth-limiting factor under high-ammonium supply is not nitrogen availability per se, but rather the secondary consequences of ammonium nutrition, including substrate acidification and cation depletion. Because these physiological responses were recorded under relatively low irradiance ($77 \mu\text{mol m}^{-2} \text{s}^{-1}$), extrapolation to higher-light commercial conditions warrants caution. The energetic and carbon costs of ammonium assimilation, and its interference with cation and cytosolic pH homeostasis, tend to intensify as nitrogen flux and growth demand rise [13, 14]; the advantage of nitrate-dominated nutrition would therefore, if anything, strengthen rather than weaken under commercial light levels, consistent with the nitrate preference for photosynthesis, biomass, leaf area and root development reported for the epiphytic orchid *Cymbidium tracyanum* [39]. Conversely, under the low irradiance applied here, the lower reductant demand of ammonium assimilation relative to nitrate [6] may have partially offset light limitation, consistent both with the intermediate apical chlorophyll optimum observed under nitrate–ammonium mixtures and with the chlorophyll increase at a moderate ammonium share reported for strawberry [32]. Apical SPAD accordingly peaked at an intermediate ratio ($\approx 45:55 \text{ NO}_3^-$ -N: NH_4^+ -N) rather than under pure nitrate, indicating that a moderate ammonium share favoured chlorophyll, a major leaf nitrogen sink, even though whole-plant growth was greatest

under nitrate. Integrated across chlorophyll fluorescence, stomatal behaviour and SPAD, nitrogen form thus modulated nitrogen investment and growth downstream of otherwise intact photosystem II photochemistry.

4.5 Temporal dynamics in the apical-basal positional interaction

The apical-basal split provides a within-treatment proxy for temporal dynamics, in which basal organs that developed during the first half of the treatment period are compared with apical organs from the second half. The strictly monopodial, indeterminate growth habit of *V. planifolia* [40] supports this approach, as each successive node is produced in a fixed acropetal sequence without branching or sympodial renewal, allowing node position to approximate a morphological time scale [41]. However, apical and basal organs inevitably differ in developmental maturity, cumulative light exposure, and proximity to assimilate sources and sinks; the relative values reported therefore integrate both treatment effects and positional gradients inherent to vine architecture and should be interpreted as indicative of spatial trends rather than isolated treatment responses at each node. Perhaps the most notable finding of this study is the significant treatment $\times$ position interaction for all five positional parameters (shoot DM, leaf DM, shoot length, total leaf area, and SPAD), indicating that nitrogen form does not merely alter total growth but reshapes the gradient of organ development along the vine. Under nitrate-dominated nutrition, a pronounced apical-basal gradient emerged, with apical shoot dry mass exceeding basal shoot dry mass by 48%. In the pure ammonium treatment, this gradient was essentially abolished. This pattern suggests that the growth-promoting effect of nitrate intensified over the 140-day treatment (though differential light interception by taller, nitrate-fed plants may have amplified this), while ammonium-related constraints accumulated to the point of eliminating the normal developmental advantage of apical parts. This finding is consistent with temporal dynamics reported for other species. In kale, 75% ammonium caused no adverse effects after four weeks but led to significant growth reductions after eight weeks [17], though no direct precedent exists for this design in orchids.

A plausible mechanism involves nitrate-dependent cytokinin biosynthesis: nitrate triggers trans-zeatin accumulation in roots, which is transported to the shoot and promotes apical meristem activity, a signalling cascade that is nitrate-specific in most species studied [42–44]. However, no hormone data were collected, and it remains unclear to what extent the apical advantage reflects nitrate signalling versus the mitigation of acidification-related constraints in nitrate-fed plants. The quadratic response of apical SPAD values is also noteworthy. The predicted optimum at approximately 45:55 NO_3^- -N: NH_4^+ -N suggests that a mixed nitrogen source may optimise chlorophyll levels, consistent with synergistic effects of combined nitrogen forms [12]. This optimum was restricted to the actively growing apical zone, as basal SPAD showed no significant response. Collectively, the positional data are consistent with the interpretation that nitrogen-form effects on *V. planifolia* intensify with prolonged exposure, likely compounded by progressive substrate acidification under ammonium-dominated nutrition. Whether this reflects genuine developmental regulation of nitrogen-form responses or simply an accumulating environmental gradient cannot be resolved from a positional proxy. Nonetheless, spatially partitioned nitrogen metabolism is documented in the CAM epiphyte *Guzmania monostachia*, whose nitrogen assimilation is functionally divided along the

leaf axis [45]; this indicates that position-dependent nitrogen-form responses are plausible in CAM species, although the developmental polarity there is not directly transferable to a monopodial vine. In rice, preferential ammonium accumulation and localised senescence in older leaves have been described as an exclusion mechanism protecting younger, photosynthetically active organs [46]; whether comparable positional partitioning contributes here cannot be tested, as apical and basal tissues were not analysed separately for nitrogen.

4.6 Practical implications

Based on the present findings, a predominantly nitrate-based nitrogen supply ($\geq 75:25$ NO_3^- -N: NH_4^+ -N) is recommended for the vegetative growth of *V. planifolia* in soilless culture. The 100:0 treatment produced the numerically highest total DM, RGR, and total leaf area among all treatments and, for total dry mass, was the only nitrogen treatment to differ significantly from both the nitrogen-free control and the pure ammonium treatment. A small ammonium supplement (up to 25%) may provide a practical compromise between maintaining high growth rates and improving chlorophyll content in apical tissue without inducing excessive substrate acidification, as the predicted SPAD optimum (54.5% NH_4^+ -N) lies at an ammonium proportion that would substantially reduce biomass accumulation and accelerate root-zone acidification. For practitioners, this recommendation translates to selecting fertilisers in which nitrate constitutes the dominant nitrogen source, a choice already standard for *Phalaenopsis* production [20] but not yet evidence-based for vanilla. These findings, established for young vegetative plants in perlite under controlled low-light conditions, provide a first evidence-based basis for nitrogen-form management in vanilla; responses at reproductive stages or under higher-irradiance commercial conditions remain to be confirmed.

These findings fill a notable gap in the vanilla nutrition literature. While previous work demonstrated that fertilisation enhances growth in soil-grown vanilla [21, 22], the present study provides the first systematic comparison of nitrogen forms for vanilla under controlled soilless conditions. The term ‘preference’ is deliberately avoided, as the growth advantage of nitrate-dominated nutrition cannot be separated from the concurrent effects of more favourable root-zone pH. Together with earlier findings on EC thresholds [23], these results contribute to an emerging evidence base for soilless vanilla production.

5 Conclusion

V. planifolia responded most favourably to nitrate-dominated nitrogen nutrition, with the 100:0 NO_3^- -N: NH_4^+ -N nutrient solution treatment yielding the highest total DM, RGR, and the numerically highest total leaf area among all treatments. Increasing ammonium proportions progressively reduced growth, acidified the root zone, and tended to reduce LMF, which remained intermediate in the pure ammonium treatment, without, however, impairing photosynthetic efficiency as measured by chlorophyll fluorescence and stomatal conductance. A distinctive treatment $\times$ position interaction revealed that nitrate nutrition selectively promoted apical vine growth, whereas ammonium flattened the apical-basal developmental gradient, providing evidence that the effects of nitrogen-form choice intensify with prolonged exposure. Relative chlorophyll content in apical leaves was optimised at an intermediate nitrogen-form ratio. Under the

controlled soilless conditions tested and recognising that nitrogen form acted together with the accompanying rhizosphere acidification rather than in isolation, these findings indicate that a predominantly nitrate-based nitrogen supply ($\geq 75:25 \text{ NO}_3^- \text{-N: NH}_4^+ \text{-N}$) is recommended for the vegetative growth phase of *V. planifolia* in soilless culture, and that nitrogen-form management has both immediate and temporally compounding effects on vanilla development.

Several limitations apply, as the experiment was confined to young vegetative plants growing over 154 days, and the absence of corrective pH adjustment means that direct and indirect effects of nitrogen form cannot be mechanistically separated. Future studies should extend to the reproductive phase and evaluate mixed nitrate–ammonium nutrition under higher irradiance, ideally in a factorial design crossing nitrogen form with light intensity, to test whether irradiance modulates the nitrogen-form response and whether the vegetative advantage of nitrate-dominated supply carries over into flowering and pod production. Including a pH-buffered or continuously corrected control would help disentangle the direct physiological effects of nitrogen form from the consequences of substrate acidification.

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Supplementary Material 4

Supplementary Material 5

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Author contributions

M. V.: Conceptualization, Methodology, Resources, Investigation, Formal Analysis, Data Curation, Visualization, Validation, Writing - Original Draft and Writing - Review & Editing. F. P.: Conceptualization, Methodology & Writing - Review & Editing. J. v. S.: Methodology & Writing - Review & Editing. A. U.: Supervision, Validation, Writing - Review & Editing, Project Administration and Funding acquisition.

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Data availability

The datasets generated during the current study are available from the corresponding author on reasonable request.

Declarations

Conflicts of interest

The authors declare no competing financial or non-financial interests.

Ethical approval

Experimental research on cultivated plants, including the collection of plant material, was carried out in accordance with all applicable institutional, national, and international guidelines and legislation throughout the course of this study.

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